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BÁO CÁO TUẦN
Thực tập tốt nghiệp - Tuần 5

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1. Yêu cầu trong tuần

- Dựa vào gợi ý về khái niệm và tính năng của Nfacct hãy xây dựng các file cấu hình của Nfacct để thêm/sửa/xoá các tài khoản. Chú ý kết hợp với iptables để kiểm tra có thực hiện được các thao tác trên hay không?
- Xây dựng API thực thi các công việc dựa vào Sequence Diagram phía dưới



2. Báo cáo công việc

a. Sử dụng nfacct

[25/12]

Thực hiện dùng lệnh nfacct help để xem các chức năng nfacct hỗ trợ

```
selunes@selunes: ~  
selunes@selunes:~$ sudo nfacct help  
[sudo] password for selunes:  
nfacct v1.0.2: utility for the Netfilter extended accounting infrastructure  
Usage: nfacct command [parameters]...  
  
Commands:  
  list [reset]          List the accounting object table (and reset)  
  add object-name       Add new accounting object to table  
  delete object-name    Delete existing accounting object  
  get object-name       Get existing accounting object  
  flush                Flush accounting object table  
  restore               Restore accounting object table reading 'list' output fr  
om stdin  
  monitor               Listens for quota attainment notifications  
  version               Display version and disclaimer  
  help                  Display this help message  
selunes@selunes:~$
```

Đôi chiếu với yêu cầu, chúng ta sẽ sử dụng các chức năng list, add, delete, flush và restore.

- Thêm accounting object mới, list accounting object

```
root@ubuntu:~# sudo nfacct list  
root@ubuntu:~# sudo nfacct list json  
root@ubuntu:~# sudo nfacct add test001  
root@ubuntu:~# sudo nfacct add test002  
root@ubuntu:~# sudo nfacct list  
{ pkts = 00000000000000000000, bytes = 00000000000000000000 } = test001;  
{ pkts = 00000000000000000000, bytes = 00000000000000000000 } = test002;  
root@ubuntu:~# sudo nfacct list json  
{ "timestamp" : 1735274051,  
  "nfacct_counters" : [  
    { "pkts" : 0, "bytes" : 0, "name" : "test001" },  
    { "pkts" : 0, "bytes" : 0, "name" : "test002" }  
  ] }
```

- Sử dụng với iptables

```
root@ubuntu:~# sudo iptables -L -v -n  
Chain INPUT (policy ACCEPT 7 packets, 1471 bytes)  
 pkts bytes target    prot opt in     out     source destination  
  
Chain FORWARD (policy ACCEPT 0 packets, 0 bytes)  
 pkts bytes target    prot opt in     out     source destination  
  
Chain OUTPUT (policy ACCEPT 7 packets, 525 bytes)  
 pkts bytes target    prot opt in     out     source destination  
root@ubuntu:~# sudo iptables -A INPUT -m nfacct --nfacct-name test001  
root@ubuntu:~# sudo iptables -A OUTPUT -m nfacct --nfacct-name test001  
root@ubuntu:~# sudo iptables -L -v | grep nfacct  
 7  995      all --    any    anywhere    anywhere    nfacct-name test001  
 8  745      all --    any    anywhere    anywhere    nfacct-name test001
```

Kết quả

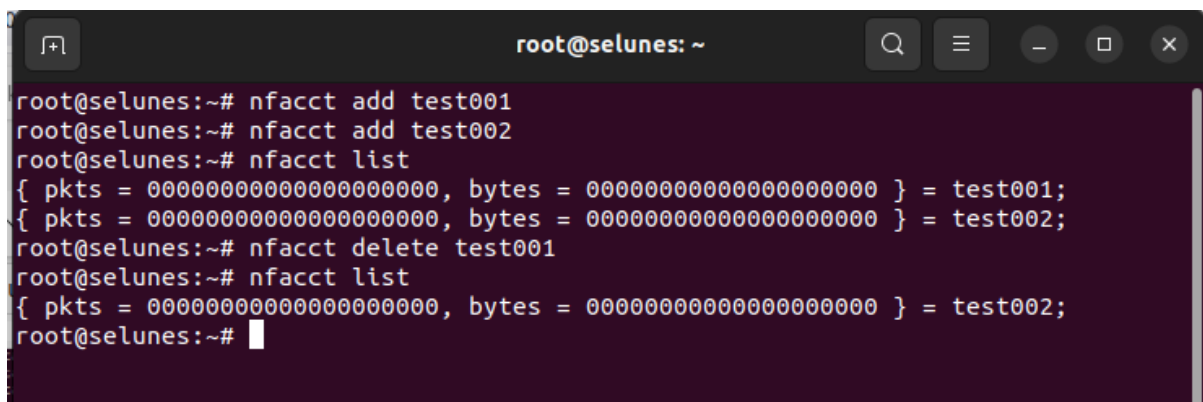
```

root@ubuntu:~# ping -c 5 10.10.16.249
PING 10.10.16.249 (10.10.16.249) 56(84) bytes of data.
64 bytes from 10.10.16.249: icmp_seq=1 ttl=128 time=0.793 ms
64 bytes from 10.10.16.249: icmp_seq=2 ttl=128 time=0.937 ms
64 bytes from 10.10.16.249: icmp_seq=3 ttl=128 time=1.24 ms
64 bytes from 10.10.16.249: icmp_seq=4 ttl=128 time=1.12 ms
64 bytes from 10.10.16.249: icmp_seq=5 ttl=128 time=1.10 ms

--- 10.10.16.249 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4008ms
rtt min/avg/max/mdev = 0.793/1.036/1.238/0.155 ms
root@ubuntu:~# sudo nfacct list
{ pkts = 00000000000000000037, bytes = 00000000000000003928 } = test001;
{ pkts = 00000000000000000000, bytes = 00000000000000000000 } = test002;

```

- Xóa accounting object

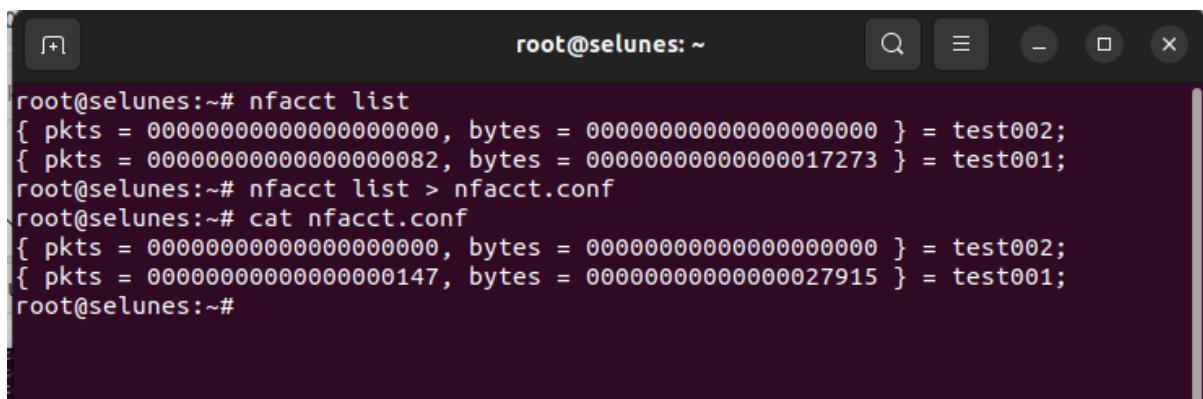


```

root@selunes:~# nfacct add test001
root@selunes:~# nfacct add test002
root@selunes:~# nfacct list
{ pkts = 00000000000000000000, bytes = 00000000000000000000 } = test001;
{ pkts = 00000000000000000000, bytes = 00000000000000000000 } = test002;
root@selunes:~# nfacct delete test001
root@selunes:~# nfacct list
{ pkts = 00000000000000000000, bytes = 00000000000000000000 } = test002;
root@selunes:~#

```

- Lưu config vào file



```

root@selunes:~# nfacct list
{ pkts = 00000000000000000000, bytes = 00000000000000000000 } = test002;
{ pkts = 00000000000000000082, bytes = 0000000000000017273 } = test001;
root@selunes:~# nfacct list > nfacct.conf
root@selunes:~# cat nfacct.conf
{ pkts = 00000000000000000000, bytes = 00000000000000000000 } = test002;
{ pkts = 00000000000000000147, bytes = 0000000000000027915 } = test001;
root@selunes:~#

```

- Load config từ file

```
root@selunes: ~
root@selunes:~# nfacct list
root@selunes:~# cat nfacct.conf
{ pkts = 00000000000000000000, bytes = 00000000000000000000 } = test002;
{ pkts = 0000000000000000000147, bytes = 00000000000000027915 } = test001;
root@selunes:~# nfacct restore < nfacct.conf
root@selunes:~# nfacct list
{ pkts = 00000000000000000000, bytes = 00000000000000000000 } = test002;
{ pkts = 0000000000000000000147, bytes = 00000000000000027915 } = test001;
root@selunes:~#
```

b. Sử dụng *libnetfilter_acct* & *libmnl*

[26/12]

Sau khi đọc code gợi ý, đối chiếu với mã nguồn của nfacct (<https://git.netfilter.org/nfacct/tree/src/nfacct.c>) phát hiện có nhiều điểm tương đồng, đồng thời đáp ứng được Sequence Diagram bên dưới, do đó nhóm thực hiện nghiên cứu file nfacct để nắm flow hoạt động, sau đó xây dựng các hàm cho phép thêm/sửa/xóa tài khoản thông qua API cung cấp bởi *libnetfilter_acct*

- List các accounting object hiện có

```
static int myGetCallback(const struct nlmsgghdr *nlh, void
*data)
{
    struct nfacct *nfacct = nfacct_alloc();

    if (nfacct == nullptr)
    {
        return MNL_CB_OK;
    }
    if (nfacct_nlmsg_parse_payload(nlh, nfacct) == -1)
    {
        nfacct_free(nfacct);
        return MNL_CB_OK;
    }

    uint64_t pkts = nfacct_attr_get_u64(nfacct,
NFACCT_ATTR_PKTS);
    uint64_t bytes = nfacct_attr_get_u64(nfacct,
NFACCT_ATTR_BYTES);
    const char *name = nfacct_attr_get_str(nfacct,
NFACCT_ATTR_NAME);
```

```

printf("Packets: %" PRIu64 "\n", ptk);
printf("Bytes: %" PRIu64 "\n", bytes);
printf("Name: %s\n", name);
printf("-----\n\n");

nfacct_free(nfacct);
return MNL_CB_OK;
}

void myGet(bool reset)
{
    struct mnl_socket *nl;
    uint32_t portId;
    nl = mnl_socket_open(NETLINK_NETFILTER);
    if (nl == nullptr)
    {
        throw std::system_error(errno, std::generic_category());
    }

    if (mnl_socket_bind(nl, 0, MNL_SOCKET_AUTOPID) < 0)
    {
        mnl_socket_close(nl);
        throw std::system_error(errno, std::generic_category());
    }

    portId = mnl_socket_get_portid(nl);
    char buf[MNL_SOCKET_BUFFER_SIZE];
    struct nlmsg_hdr *nlh;
    unsigned int seq = time(NULL);
    int ret;

    nlh = nfacct_nlmsg_build_hdr(buf, reset ?
    NFNL_MSG_ACCT_GET_CTRZERO : NFNL_MSG_ACCT_GET,
    NLM_F_DUMP, seq);
    if (mnl_socket_sendto(nl, nlh, nlh->nmsg_len) == -1)
    {

```

```

throw std::system_error(errno, std::generic_category());
}
while ((ret = mnl_socket_recvfrom(nl, buf, sizeof(buf)))
> 0)
{
if ((ret = mnl_cb_run(buf, ret, seq, portId,
myGetCallback, NULL)) <= 0)
break;
}

if (ret == -1)
{
throw std::system_error(errno, std::generic_category());
}
mnl_socket_close(nl);
}

```

Kết quả:

```

Choose an option:
1. Add a new account (myAdd)
2. Get account information (myGet)
3. Delete an account (myDelete)
4. Exit
Enter your choice: 2
Getting account information...
Packets: 0
Bytes: 0
Name: test002
-----

Packets: 147
Bytes: 27915
Name: test001
-----

```

- Thêm accounting object

```

void myAdd(const char *name)
{
struct mnl_socket *nl;
uint32_t portId;
nl = mnl_socket_open(NETLINK_NETFILTER);

```

```

if (nl == nullptr)
{
throw std::system_error(errno, std::generic_category());
}

if (mnl_socket_bind(nl, 0, MNL_SOCKET_AUTOPID) < 0)
{
mnl_socket_close(nl);
throw std::system_error(errno, std::generic_category());
}

portId = mnl_socket_get_portid(nl);
uint64_t pkts = 0;
uint64_t bytes = 0;
uint32_t seq = time(NULL);
int ret;
struct nfacct *nfacct = nfacct_alloc();
struct nlmsg_hdr *nlh;
char buf[MNL_SOCKET_BUFFER_SIZE];

if (nfacct == nullptr)
{
throw std::system_error(errno, std::generic_category());
}

nfacct_attr_set(nfacct, NFACCT_ATTR_NAME, name);
nfacct_attr_set_u64(nfacct, NFACCT_ATTR_PKTS, pkts);
nfacct_attr_set_u64(nfacct, NFACCT_ATTR_BYTES, bytes);

nlh = nfacct_nlmsg_build_hdr(buf, NFNL_MSG_ACCT_NEW,
NLM_F_CREATE | NLM_F_ACK, seq);
nfacct_nlmsg_build_payload(nlh, nfacct);
nfacct_free(nfacct);

if (mnl_socket_sendto(nl, nlh, nlh->nlmsg_len) == -1)
{
throw std::system_error(errno, std::generic_category());
}
while (ret > 0)

```



```
{  
ret = mnl_cb_run(buf, ret, seq, portId, NULL, NULL);  
if (ret <= 0)  
break;  
ret = mnl_socket_recvfrom(nl, buf, sizeof(buf));  
}  
if (ret == -1)  
{  
throw std::system_error(errno, std::generic_category());  
}  
mnl_socket_close(nl);  
}
```

Kết quả:

```
Choose an option:
1. Add a new account (myAdd)
2. Get account information (myGet)
3. Delete an account (myDelete)
4. Exit
Enter your choice: 1
Enter the name for the new account: test003
Account added successfully!
Choose an option:
```

```
Choose an option:
1. Add a new account (myAdd)
2. Get account information (myGet)
3. Delete an account (myDelete)
4. Exit
Enter your choice: 2
Getting account information...
Packets: 0
Bytes: 0
Name: test002
-----

Packets: 147
Bytes: 27915
Name: test001
-----

Packets: 0
Bytes: 0
Name: test003
-----
```

- Xóa accounting object

```
void myDelete(const char *name)
{
    struct mnl_socket *nl;
    uint32_t portId;
    nl = mnl_socket_open(NETLINK_NETFILTER);
    if (nl == nullptr)
    {
        throw std::system_error(errno, std::generic_category());
    }

    if (mnl_socket_bind(nl, 0, MNL_SOCKET_AUTOPID) < 0)
    {
        mnl_socket_close(nl);
        throw std::system_error(errno, std::generic_category());
    }
}
```

```

}

portId = mnl_socket_get_portid(nl);
uint32_t seq = time(NULL);
int ret;
struct nfacct *nfacct = nfacct_alloc();
struct nlmsg_hdr *nlh;
char buf[MNL_SOCKET_BUFFER_SIZE];

if (nfacct == nullptr)
{
    throw std::system_error(errno, std::generic_category());
}

nfacct_attr_set(nfacct, NFACCT_ATTR_NAME, name);

nlh = nfacct_nlmsg_build_hdr(buf, NFNL_MSG_ACCT_DEL,
NLM_F_ACK, seq);
nfacct_nlmsg_build_payload(nlh, nfacct);
nfacct_free(nfacct);

if (mnl_socket_sendto(nl, nlh, nlh->nlmsg_len) == -1)
{
    throw std::system_error(errno, std::generic_category());
}

while (ret > 0)
{
    ret = mnl_cb_run(buf, ret, seq, portId, NULL, NULL);
    if (ret <= 0)
        break;
    ret = mnl_socket_recvfrom(nl, buf, sizeof(buf));
}
if (ret == -1)
{
    throw std::system_error(errno, std::generic_category());
}
mnl_socket_close(nl);

```

```
}
```

Kết quả

```
Choose an option:
1. Add a new account (myAdd)
2. Get account information (myGet)
3. Delete an account (myDelete)
4. Exit
Enter your choice: 3
Enter the name of the account to delete: test003
Account deleted successfully!
```

```
Choose an option:
1. Add a new account (myAdd)
2. Get account information (myGet)
3. Delete an account (myDelete)
4. Exit
Enter your choice: 2
Getting account information...
Packets: 0
Bytes: 0
Name: test002
-----

Packets: 147
Bytes: 27915
Name: test001
-----
```